main.py | Output

```
6  def rotate_left_without_carry(binary, positions):
7      print("\n--- Rotate Left Without Carry (ROL) ---")
8      print("Original binary:", binary)
9
10     result = binary
11
12     for i in range(positions):
13         msb = result[0]
14         result = result[1:] + msb
15
16         print("Step", i + 1, ":", result)
17
18     print("Final 8-bit result:", result)
19
20
21 def rotate_left_with_carry(binary, positions, carry):
22     print("\n--- Rotate Left With Carry (RCL) ---")
23     print("Original binary:", binary)
24     print("Initial carry flag:", carry)
25
```

Previous code was kept. Restore previous code ×

```
28     for i in range(positions):
```

main.py | Output

```
26      result = binary
27
28      for i in range(positions):
29          msb = result[0]
30
31          # Shift left and insert
    the old carry at the right
32          result = result[1:] +
    str(carry)
33
34          # Old MSB becomes the new
    carry
35          carry = int(msb)
36
37          print("Step", i + 1, ":",
    result, "| Carry =", carry)
38
39      print("Final 8-bit result:",
    result)
40      print("Final carry flag:",
    carry)
41
42
43  # Given binary number
44  binary_number = "10110110"
45
46  # (a) ROL by 2 positions
47  rotate_left_without_carry(binary_nu
    mber, 2)
48
50  rotate_left_with_carry(binary_numbe
```

Previous code was kept. Restore previous code ×

main.py | Output

```
    for i in range(positions):
        msb = result[0]

        # Shift left and insert the old carry at the right
        result = result[1:] + str(carry)

        # Old MSB becomes the new carry
        carry = int(msb)

        print("Step", i + 1, ":", result, "| Carry =", carry)

    print("Final 8-bit result:", result)
    print("Final carry flag:", carry)


# Given binary number
binary_number = "10110110"

# (a) ROL by 2 positions
rotate_left_without_carry(binary_number, 2)

# (b) RCL by 2 positions, assuming initial carry = 0
rotate_left_with_carry(binary_number, 2, 0)
```

Visualize | Run

main.py Output

```
--- Rotate Left Without Carry (ROL) --
-
Original binary: 10110110
Step 1 : 01101101
Step 2 : 11011010
Final 8-bit result: 11011010

--- Rotate Left With Carry (RCL) ---
Original binary: 10110110
Initial carry flag: 0
Step 1 : 01101100 | Carry = 1
Step 2 : 11011001 | Carry = 0
Final 8-bit result: 11011001
Final carry flag: 0
```

Previous code was kept. Restore previous code ×